Beyond the Pitch Accent

A floating tone reanalysis of Tokyo Japanese postlexical tonology

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Abstract

This paper offers a floating tone reanalysis of Tokyo Japanese postlexical tonology. Existing analyses cannot reconcile the tonal behaviour of recessive particles after nominal versus non-nominal bases. I argue that nouns and *na*-adjective stems underlyingly bear a trailing floating tone, which overrides the particle-initial tone in recessive cliticization, formalized as the Nominal Overriding Generalization. This analysis derives the clitic and phrasal patterns, including the adverbial usage effect, without ad hoc stipulations. Historical evidence shows how floating tones arose from regular changes and analogy. Tokyo's surface pitch accent is thus underlain by level tones.

Keywords: Tokyo Japanese; pitch accent; floating tone; clitic

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0 Introduction

Tokyo Japanese has long been taken as the textbook exemplification of pitch accent systems. The underlying (n+1)-way tonal contrast in n-syllabic words is traditionally analysed as a high-toned accent on, or a downstep after, any of the n syllables, in addition to an unaccented pattern. The present paper argues that the standard analysis, like other existing analyses, makes too many stipulations in accounting for postlexical tonological patterns. A reanalysis is therefore called for, based on the following assumption:

- In the postlexical tonology of Tokyo Japanese, at the end of nominal (= noun and *na*-adjective) bases, there is an obligatory "floating skeleton" that requires a floating tone, either high or low, capable of overriding the underlying tonal specification of a following particle.

This paper is primarily dedicated to the clitic and phrasal tonology of Tokyo Japanese; its lexical tonology is addressed only in passing when relevant.

Uncontroversial Modern Japanese data are not cited. The reader may easily verify the data against the entries and accentuation rules in the 3 authoritative pitch accent dictionaries—Hirayama (1960), Akinaga (2014), and NHK (2016). When historical Japanese tonology is relevant, Ramsey's reinterpretation (Ramsey 1979; de Boer 2010) is adopted.

This paper is organized as follows. § 1 introduces basic facts about Tokyo phonology and morphology. § 2 is a critical review of existing theories of Tokyo postlexical tonology. § 3 is my floating tone reanalysis of the pattern. § 4 addresses the origin of floating tones from a historical perspective. § 5 discusses a few theoretical issues related to the reanalysis. § 6 concludes this paper.

0.1 Conventions

- Representations:
  - /*xxx*/: representation at clitic input (UR, "Underlying Representation").
  - [*xxx*]: representation at clitic output (SR, "Surface Representation").
  - ⟦*xxx*⟧: representation at phrasal output (PR, "Phrasal/Phonetic Representation").
  - ⟨*xxx*⟩ is used to denote linguistic signs without touching upon their accentuation or specifying which level the representation is on.
- Transcription:
  - Capitals (*A*, *E*, *I*, *N*, *O*, *Q*, *U*): defective moras.
  - *x*ˈ*x*: downstep.
  - *xx*° is used to emphasize the absence of a downstep when necessary.
- Mora-tone symbols:
  - ❶: high-toned mora (undownstepped).
  - Ⓛ: low-toned mora (downstepped once or more).
  - ◍: high- or low-toned mora; mora unspecified for tone; devoiced mora without an identifiable tone.
  - ▼: defective ❶ (usually potentially subject to Defective Lowering).
  - ▽: defective Ⓛ (usually lowered through Defective Lowering).
  - ❹: nondefective ❶.
- Lexemes:
  - ⟨*xxx*^{i}⟩: *i*-adjective (lemma = indicative = adnominal = ⟨*xxx*-*I*⟩).
  - ⟨*xxx*^{na}⟩: *na*-adjective (lemma = ∅-indicative = ⟨*xxx*⟩, *da*-indicative = ⟨*xxx*-*da*⟩, adnominal = ⟨*xxx*-*na*⟩).
  - ⟨*xxc*^{v}⟩: athematic verb (lemma = indicative = adnominal = ⟨*xxc*-*u*⟩; *c* = any consonant).

    - ⟨*xxi*^{v}⟩, ⟨*xxe*^{v}⟩: *i*-, *e*-thematic verb (lemma = indicative = adnominal = ⟨*xxi-ru*⟩, ⟨*xxe-ru*⟩).
- Boundaries:
    - -: subword boundary.
    - =: cliticization boundary.
- Relations:
    - >: regular sound change.
    - ≫: analogical change.
    - →: phonological derivation.
    - ⇒: inference.
    - ~: variation.

1 Basic facts about Tokyo Japanese

1.1 Subphrasal phonology

The surface tonology of Tokyo Japanese within a tonological word (henceforth **t-word**)[1] is characterized by whether there is a phonemic pitch fall—**downstep**—and, if so, where it is located in the t-word. With the surface downstep always appearing between two moras, the maximal n-way[2] surface tonal contrast on an n-mora t-word is illustrated as in (1):

(1) The 4-way surface tonal contrast in 4-mora t-words

PR		gloss	
downstep	mora-tone		
ba'nana=o	❶Ⓛ Ⓛ=Ⓛ	"banana=ACC"	with a downstep
tama'go=o	❶❶Ⓛ=Ⓛ	"egg=ACC"	
sasimi'=o	❶❶❶=Ⓛ	"sashimi=ACC"	
itigo=o	❶❶❶=❶	"strawberry=ACC"	with no downstep

Note that there's no phonemic pitch rise. This surface pattern can be stated as the following constraint:

- **Monotonicity Constraint**: A t-word has a tonal melody of [❶$_1$Ⓛ$_0$] (in *SPE* notation).

The maximal n-way phonetic contrast on an n-mora word is reduced if some of the moras are defective. Mora defectiveness in this context can be defined in terms of syllabicity: the first mora of a syllable nucleus is **nondefective**, whereas all remaining moras are **defective**[3]. The two mora types are exemplified in (2):

(2) Nondefective and defective moras (dot (.): syllable boundary; capital letter (*U/O/I/N/Q/...*): defective mora)

mora type	structure	example	gloss
nondefective	(C)(*y*)V	*o.mo.tya.ya*[4]	"toy store"
defective	vowel lengthening	*zyuU.yoO*	"to be important"
	offglide	*seI.kaI*	"correct answer"
	nasal	*siN.buN*	"newspaper"
	obstruent geminate	*iQ.sa.tu*	"one (of books)"

The constraint on the phonetic position of the downstep can be stated statically:

- **Defectiveness Constraint**: A downstep never follows a defective mora.

This constraint is actually the direct consequence of a repair rule (see Akinaga 1968):

- **Defective Lowering**: Whenever a downstep would otherwise follow a defective mora, the downstep is shifted to the left of this mora.

  Loanword accentuation of Tokyo Japanese illustrates this well. Normally the downstep comes just before

[1] The t-word is defined on the grounds of tonological evidence, and may be effectively similar to the concept of the prosodic/phonological word. Whether these different criteria precisely converge in Tokyo Japanese is beyond the scope of this paper, though. Similar for the **t-phrase**, for which see below.

[2] Rather than (n+1)-way, as many texts have often described. See § 5.2.

[3] Or "special" (*tokusyu-na*).

[4] In the context of Japanese phonology, it is customary to parse a syllable exhaustively into moras, rather than leaving onsets behind. The "mora" in this sense is rather a direct translation of the traditional term *haku* (lit. "beat").

the last two moras[5] (Akinaga 2014: #9), as in (3a). However, a defective mora preceding this underlying downstep is subject to Defective Lowering, yielding a left-shifted surface downstep, as in (3b).

(3) Loanword accentuation and Defective Lowering

(a)	*kurisuˈmasu*	…❶❶ⓁⓁ			“Christmas”	default
	bitaˈmiN				“vitamin”	
	pasupoˈOto				“passport”	
(b)	*paInaQˈpuru*	…❶❶ⓁⓁ	→	*paInaˈQpuru*	…❶ⓁⓁⓁ	“pineapple”	with Defective Lowering
	haNbaAˈgaA			*haNbaˈAgaA*		“hamburger”	
	wasiNˈtoN			*wasiˈNtoN*		“Washington”	

For cases like (3b), I will henceforth write /…❶▼ⓁⓁ/ → […❶▽ⓁⓁ] instead, with iconic down-pointing triangles for tones (potentially) subject to Defective Lowering, to emphasize the syllabicity condition of the process.

1.2 Morphology

There are 6 major lexical parts of speech (henceforth PoSs) in Japanese: **verbs**, ***i*-adjectives**, ***na*-adjectives**, **nouns**, **adverbs**, and **adnominals**. Among them, verbs and adjectives are inflectable. Verbs, according to their segmental conjugation types, can be further divided into thematic and athematic. In addition, nouns and *na*-adjectives are collectively called **nominals** in this paper, as they are historically related and behave similarly in synchronic tonology (see § 3.1).

In postlexical tonology, phonologically independent PoSs have fixed (rather than movable) accents, as shown in (4). This means that the downstep either stays unshifted as in the base or is supplied by a **recessive** particle, unless it is overridden by a **dominant** particle.

(4) Noun patterns, recessive and dominant

	unaccented	accented			
		final-accented	medial-accented	initial-accented	
	⟨*itigo*⟩ “strawberry”	⟨*sasimi*⟩ “sashimi”	⟨*tamago*⟩ “egg”	⟨*banana*⟩ “banana”	
[…]	❶❶❶	❶❶❶	❶❶Ⓛ	❶ⓁⓁ	recessive
[…=*hodo*]	❶❶❶=❶❶	❶❶❶=ⓁⓁ	❶❶Ⓛ=ⓁⓁ	❶ⓁⓁ=ⓁⓁ	
[…=*nara*]	❶❶❶=❶Ⓛ	❶❶❶=ⓁⓁ	❶❶Ⓛ=ⓁⓁ	❶ⓁⓁ=ⓁⓁ	
[…=*dake*]	❶❶❶=❶❶	❶❶❶=❶❶	❶❶❶=❶❶	❶❶❶=❶❶	dominant
[…=*rasiku*]	❶❶❶=❶ⓁⓁ	❶❶❶=❶ⓁⓁ	❶❶❶=❶ⓁⓁ	❶❶❶=❶ⓁⓁ	

The base is said to be **accented** if it contributes a downstep (either within the base or at the juncture) when combined with a recessive particle, and **unaccented** if not. For an accented base, it is **initial-accented** when the downstep follows the first mora of the base, **final-accented** when the downstep follows the last mora of the base, or otherwise **medial-accented**.

Largely the same patterns apply to verb and *i*-adjective bases, as in (5). The only difference is that for verbs and *i*-adjectives, there are no tonal subclasses for accented lexemes. For the lemma and the infinitive—the commonest wordforms of verbs and *i*-adjectives as cliticization bases—the downstep usually comes just

[5] One might argue to dismiss moras as a useful concept in the discussions of Tokyo tonology, given that downstep opportunity corresponds well to the syllable count. Loanword accentuation shows this is not the case, because it would otherwise prevent us from making a generalization about the accentuation pattern in (3).

before the last mora of the wordform.

(5) Verb and *i*-adjective patterns, recessive and dominant

	verb lemma, unaccented	*i*-adj. lemma, unaccented	verb lemma, accented	*i*-adj. lemma, accented	
	⟨*arau*⟩ "to wash"	⟨*akaI*⟩ "to be red"	⟨*taberu*⟩ "to eat"	⟨*siroI*⟩ "to be white"	
[…]	❶❶❶	❶❶❶	❶❶Ⓛ	❶❶Ⓛ	recessive
[…=*hodo*]	❶❶❶=❶❶	❶❶❶=❶❶	❶❶Ⓛ=ⓁⓁ	❶❶Ⓛ=ⓁⓁ	
[…=*nara*]	❶❶❶=ⓁⓁ	❶❶▽=ⓁⓁ[6]	❶❶Ⓛ=ⓁⓁ	❶❶Ⓛ=ⓁⓁ	
[…=*dake*]	❶❶❶=❶❶	❶❶❶=❶❶	❶❶❶=❶❶	❶❶❶=❶❶	dominant
[…=*rasiku*]	❶❶❶=❶ⓁⓁ	❶❶❶=❶ⓁⓁ	❶❶❶=❶ⓁⓁ	❶❶❶=❶ⓁⓁ	

Of the two major patterns, the dominant pattern is less interesting, because the resulting accentuation simply disregards the tonal information of the base. In contrast, the recessive pattern, being the more productive type, accounts for the widest range of tonal cliticization (henceforth **t-cliticization**), and largely resembles tonal phrasing (henceforth **t-phrasing**).

1.3 Phrasal phonology

1.3.1 T-phrasing: general t-phrases

As shown in (6), the phrasal tonology of Tokyo Japanese is a full-fledged downstep system, where there can be zero or more downsteps in a tonological phrase (henceforth **t-phrase**)[7]. T-phrases are built through t-phrasing, whose first approximation is given below:

- **T-phrasing generalization** (first approximation, covering only general t-phrases): The downsteps in a t-phrase are directly inherited from those in the PRs of its building t-words.

This first approximation applies to cases where the element directly before the juncture is not a noun, and such a phrase is provisionally called a **general t-phrase** here.

(6) Phrasal tonology of N=P + V (**undownstepped baseline**; Initial Lowering (see § 1.3.2))

	visualized pitch	PR		visualized pitch	PR	
		downstep	mora-tone		downstep	mora-tone
				ta ***be*** *ru*	*tabeꜜru*	❶❶Ⓛ
(a)	***ba*** *nana=o*	*baꜜnana=o*	❶ⓁⓁ=Ⓛ	***ba*** *nana=o‿tabe* *ru*	*baꜜnana=o‿tabeꜜru*	❶ⓁⓁ=Ⓛ‿ⓁⓁⓁ
(b)	*ta* ***ma*** *go=o*	*tamaꜜgo=o*	❶❶Ⓛ=Ⓛ	*ta* ***ma*** *go=o‿tabe* *ru*	*tamaꜜgo=o‿tabeꜜru*	❶❶Ⓛ=Ⓛ‿ⓁⓁⓁ
(c)	*sa* ***simi*** *=o*	*sasimiꜜ=o*	❶❶❶=Ⓛ	*sa* ***simi*** *=o‿tabe* *ru*	*sasimiꜜ=o‿tabeꜜru*	❶❶❶=Ⓛ‿ⓁⓁⓁ
(d)	*i* ***tigo=o***	*itigo=o*	❶❶❶=❶	*i* ***tigo=o‿tabe*** *ru*	*itigo=o‿tabeꜜru*	❶❶❶=❶‿❶❶❶
	"banana/egg/sashimi/strawberry"			"eat"; "eat banana/egg/sashimi/strawberry"		

1.3.2 Structure of a t-phrase

Within a t-phrase, the pitch drop of the first downstep is the steepest, and all pitch drops after that are

[6] ❶❶▽=ⓁⓁ ← ❶❶▼=ⓁⓁ, through Defective Lowering.

[7] See footnote 1.

significantly reduced if still preserved. It is thus customary in some sources (including Hirayama 1960; Akinaga 2014) to omit all downsteps from the second onwards within a t-phrase. This convention is also embodied in my mora-tone notation for PRs, where in (6abc) twice downstepped *ru* is given the same symbol Ⓒ as once downstepped *tabe*.

The first mora of a t-phrase, if not followed by a downstep, is lowered compared with the following mora.[8] For example, the first mora of ⟦*tama*ꜜ*go_tabe*ꜜ*ru*⟧ in (6b) is lowered, whereas the first mora of ⟦***ba***ꜜ*nana_tabe*ꜜ*ru*⟧ in (6a) remains unaffected. This automatic process, **Initial Lowering**, is systematically abstracted away from my downstep and mora-tone notations.[9]

1.3.3 T-phrasing: bare-argumental t-phrases

The first approximation of t-phrasing doesn't directly apply to N+X t-phrases, in which the word juncture is preceded by a bare noun (i.e., without trailing particles). One major source of this structure is **bare-argumental t-phrases**. In Japanese, argument-marking particles can be omitted under certain conditions.[10] In that case, if the bare argument is t-phrased with the following predicate, a slightly less transparent pattern is observed:[11]

- **T-phrasing generalization** (final version, covering both general t-phrases and bare-argumental t-phrases): The downsteps in a t-phrase are directly inherited from those in the PRs of its building t-words; there's also an additional downstep after a final-accented noun argument.

(7) Bare-argumental t-phrase vs. general t-phrase

	⟦… + *tabeꜜru*⟧ "eat …" (bare-arg'l t-phrase)	⟦…=ACC + *tabeꜜru*⟧ "eat …" (general t-phrase)	PR of the bare argument in isolation		
(a)	*baꜜnana_tabeꜜru*	*baꜜnana=o_tabeꜜru*	*baꜜnana*	"banana"	initial-accented
(b)	*tamaꜜgo_tabeꜜru*	*tamaꜜgo=o_tabeꜜru*	*tamaꜜgo*	"egg"	medial-accented
(c)	*sasimiꜜ_tabeꜜru*	*sasimiꜜ=o_tabeꜜru*	*sasimi*	"sashimi"	final-accented
(d)	*itigo_tabeꜜru*	*itigo=o_tabeꜜru*	*itigo*	"strawberry"	unaccented

In (7c), with ⟦*sasimi*⟧ and ⟦*tabeꜜru*⟧ being the PRs of the building t-words in isolation, we would otherwise get the wrong result ˣ⟦*sasimi_tabeꜜru*⟧ if we simply concatenated the PRs, as we assumed in the first approximation of t-phrasing. Instead, the final version of t-phrasing needs to "recover the underlying downstep from ⟦*sasimiꜜ=o*⟧" to yield the correct result ⟦*sasimiꜜ_tabeꜜru*⟧.[12]

Note especially that bare nouns serving adverbial usage follow a different subphrasal pattern, for which see § 2.7 and § 3.5.2.[13]

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[8] Although this lowering is considered optional and stiff when the initial mora is followed by a sonorant defective mora. (Akinaga 2014: FM p. 8; NHK 2016: app. p. 11; Poser 1984: 195)

[9] Note that Initial Lowering obscures the monotonicity constraint.

[10] Specifically, the nominative ⟨=*ga*⟩, the accusative ⟨=*o*⟩, and, for motion verbs, the goal-marking dative ⟨=*ni*⟩.

[11] The earliest description of the bare-argumental t-phrase pattern is attributed to Kawakami (1963) according to Kori (2015), and this pattern is confirmed in Akinaga (1966: 84; 2014: #98).

[12] Alternatively, the bare-argumental t-phrase pattern (⟦*sasimiꜜ_tabeꜜru*⟧) can be viewed as phonetically eliding the omitted particle (⟨=*o*⟩) directly from the PR of its general t-phrase counterpart (⟦*sasimiꜜ=o_tabeꜜru*⟧), leaving the accentuation intact.

[13] The adverbial usage effect and the bare-argumental t-phrase pattern are competing. As Akinaga (2014: #62) notes the tendency to disregard the adverbial usage effect in innovative Tokyo, we may further confirm that, of the two competing phenomena, the adverbial usage effect is the morphophonologically marked one, whereas the bare-argumental t-phrase pattern is the default one synchronically.

1.3.4 Vowel Devoicing and downstep shift

Tokyo Japanese vowels are devoiced under certain conditions, typically when a short close vowel (*i* or *u*) is flanked by a pair of voiceless consonants. Here we are concerned with two cases where a devoiced mora is followed by a downstep: (Akinaga 2014: #c)

- When the devoiced mora is not initial, there is said to be an optional leftward downstep shift: ⟦***itigo***=***si̥***ꜜ*ka* ❶❶❶=◍ꜜⓁ⟧ "except for strawberry" is analysed by some as ⟦***itigo***=ꜜ*si̥ka* ❶❶❶=ꜜ◍Ⓛ⟧, with a leftward downstep shift, because it makes little phonetic difference whether the devoiced mora *si̥* is specified as ❶ or Ⓛ.
- When the devoiced mora is initial, there is said to be an optional rightward downstep shift: ⟦*hu̥*ꜜ*ku*‿*kaze* ◍ꜜⓁ‿ⓁⓁ⟧ "blowing wind" is optionally pronounced and analysed by some as ⟦*hu̥****ku***ꜜ‿*kaze* ◍❶ꜜ‿ⓁⓁ⟧.

Vowel Devoicing and downstep shifts induced thereby, as low-level variations in phonetic realization, are generally abstracted away in this paper.

2 Tokyo postlexical tonology in existing analyses

This section is a comparative survey of major existing analyses of Tokyo postlexical tonology. In comparing and evaluating these theories, I will abstract away their formal details and concentrate on their empirical consequences. Specifically, the underlying contrast in Tokyo tonology has been variously modelled as a downstep at mora/syllable boundaries, or a concrete high tone / a purely diacritical accent on the mora/syllable before the downstep. In this section, I will uniformly translate them into statements about the existence and positioning of downsteps, whenever such a simplification is empirically equivalent for the present purpose. This translation is made possible by the fact that, within any single existing theory, the structure of the assumed "accent" in the postlexical tonology of Tokyo stays invariant across all lexical classes and items.

2.1 Standard analysis: aligning URs with bare-argumental t-phrases

The standard analysis, represented by Akinaga (1966; 2014), is widely assumed by both native (notably Hirayama 1960; NHK 2016; Haraguchi 1975; Kubozono 1987) and foreign (notably Bloch 1946; Martin 1952; Poser 1984; Vance 1987) Japanese phonologists. In this analysis, the UR of a t-word differs from the PR only when the t-word is a final-accented nominal (= noun or *na*-adjective), in having an additional downstep at the end of the UR, as exemplified in (8):

(8) Comparison of the standard URs and PRs of t-words

standard UR	PR	last element of the t-word	
itigo=o		particle	
akaI		unaccented *i*-adjective	
noru		unaccented verb	final-[❶]
itigo		unaccented nominal	
sasimiˈ	*sasimi*	final-accented nominal	
baˈnana		accented nominal	
tamaˈgo			
kaˈku		accented verb	final-[Ⓛ]
siroˈI		accented *i*-adjective	
sasimiˈ=o		particle	

Such URs are assumed so that t-phrasing can be derived through simple concatenation of URs, as shown in (9):

(9) Final-[❶] t-words and phrases led thereby

	final-[❶] t-word		phrase led by the t-word		
	last element	standard UR	PR	gloss	
(a)	final-acc. noun	*sasimiˈ*	*sasimiˈ_tabeˈru*	"eat sashimi" (sashimi + eat)	bare-arg'l
(b)	unacc. noun	*itigo°*	*itigo_tabeˈru*	"eat strawb" (strawb + eat)	t-phrase
(c)	particle	*itigo=o°*	*itigo=o_tabeˈru*	"eat strawb" (strawb=ACC + eat)	general t-phrase
(d)	verb	*arau°*	*arau_itigo*	"strawb to wash" (wash + strawb)	
(e)	*i*-adjective	*akaI°*	*akaI_itigo*	"red strawb" (red + strawb)	

These phrasal patterns also align well with the pattern of some recessive particles like ⟨=*hodo*⟩, where the particle always surfaces as initial-[❶] after a final-[❶] non-nominal, as in (10bcd): (Kori 2020)

(10) Final-[❶] base + ⟨=*hodo*⟩ (the standard analysis)

	[...=*hodo*] "as much as"	final-[❶] base		
		standard UR	gloss	last element
(a)	❶❶❶=❶❶	*itigo*°	"strawb"	unaccented noun
(b)	❶❶❶=❶❶	*arau*°	"wash"	verb
(c)	❶❶❶=❶❶	*akaI*°	"be red"	*i*-adjective
(d)	❶❶❶=❶=❶❶	*itigo=ni*°	"strawb=DAT"	particle
(e)	❶❶❶=Ⓛ Ⓛ	*sasimiˈ*	"sashimi"	final-accented noun

2.2 Problem with the standard analysis: ⟨...=*nara*⟩

There is, however, a major problem with the standard analysis in accounting for the most frequent recessive pattern, represented by ⟨=*nara*⟩:

(11) Final-[❶] base + ⟨=*nara*⟩ in the standard analysis (problematic cases)

	[...=*nara*] "if ..."	final-[❶] base			
		standard UR	gloss	last element	
(a)	❶❶❶=❶Ⓛ	*itigo*°	"strawb"	unaccented noun	⇒ /=*naˈra*/
(b)	❶❶❶=Ⓛ Ⓛ	*arau*°	"wash"	verb	⇒ either simply /=ˈ*nara*/, or /=*naˈra*/ with downstep insertion at the juncture
(c)	❶❶▼=Ⓛ Ⓛ → ❶❶▽=Ⓛ Ⓛ	*akaI*°	"be red"	*i*-adjective	
(d)	❶❶❶=❶=Ⓛ Ⓛ	*itigo=ni*°	"strawb=DAT"	particle	
(e)	❶❶❶=Ⓛ Ⓛ	*sasimiˈ*	"sashimi"	final-accented noun	

As shown in (11bcd), ⟨=*nara*⟩ is [=Ⓛ Ⓛ] after final-[❶] non-nominal bases, which don't have a trailing downstep in the standard analysis. However, this [=Ⓛ Ⓛ] realization surprisingly patterns with the realization after final-accented nominals (11e), to the exclusion of unaccented nominals (11a). (Akinaga 1966; 2014; Kori 2020)

⟨=*nara*⟩'s behaviour cannot be dismissed as idiosyncratic. According to the summary list compiled by Kori (2020), recessive particles that show this tonal mismatch after nominals vs. non-nominals, like ⟨=*nara*⟩, constitute the majority of the list.

Unfortunately, the standard analysis, as represented by Akinaga (1966; 2014), does nothing more than listing the nominal vs. non-nominal patterns separately, implying a split in the UR of the particle (say, /=*naˈra*/ after a nominal and /=ˈ*nara*/ (or /=ˈ*naˈra*/) after other PoSs). Alternatively (as in Martin 1952: § 7.75[14]), this pattern can be analysed as downstep insertion between the base and the particle, which have a uniform UR like /=*naˈra*/ or so. However, the downstep insertion and its condition (applying only to ⟨=*nara*⟩-type particles but not to ⟨=*hodo*⟩-type particles) are again stipulated rather than independently motivated.

2.3 Martin's reanalysis: aligning URs with ⟨...=*nara*⟩

Martin's (1967; 1975) later reanalysis, unlike the standard analysis (a version of which was adopted by the same author, Martin (1952)), is instead based on the pattern when the base is followed by ⟨=*nara*⟩-type particles:

[14] Note that Martin at that time worked with a version of the standard analysis, instead of his (1967; 1975) reanalysis, to which we shortly turn.

(12) Final-[❶] base + ⟨=*nara*⟩ in Martin's analysis

	[...=*nara*] "if ..."	final-[❶] base		
		Martin's UR	gloss	last element
(a)	❶❶❶=❶Ⓛ	*itigo*°	"strawb"	unaccented noun
(b)	❶❶❶=ⓁⓁ	*arau*ꜜ	"wash"	verb
(c)	❶❶▼=ⓁⓁ → ❶❶▽=ⓁⓁ	*akaI*ꜜ	"be red"	*i*-adjective
(d)	❶❶❶=❶=ⓁⓁ	*itigo=ni*ꜜ	"strawb=DAT"	particle
(e)	❶❶❶=ⓁⓁ	*sasimi*ꜜ	"sashimi"	final-accented noun

Martin in his reanalysis attributes the intervening downsteps in (12bcd) to the URs of the preceding final-[❶] non-nominals. Therefore, ⟨=*nara*⟩-type particles no longer show a split tonology; a uniform UR of /=*na*ꜜ*ra*/ can be assumed after bases of all PoSs.

Largely the same analysis is followed by Sato (1999).

2.4 Problem with Martin's reanalysis: bare-argumental t-phrases and ⟨...=*hodo*⟩

Unfortunately, Martin's reanalysis creates a problem for bare-argumental t-phrasing patterns.

(13) Final-[❶] t-words and phrases led thereby: comparison of the standard analysis and Martin's reanalysis (problematic cases for Martin's reanalysis)

	final-[❶] t-word			phrase led by the t-word		
	Martin's UR	standard UR	PR	PR	gloss	
(a)	*sasimi*ꜜ		*sasimi*	*sasimiꜜ_arau*	"wash sashimi" (sashimi + wash)	bare-arg'l t-phrase
(b)	*itigo*			*itigo_arau*	"wash strawb" (strawb + wash)	
(c)	*itigo=o*ꜜ	*itigo=o*		*itigo=o_arau*	"wash strawb" (strawb=ACC + wash)	general t-phrase
(d)	*arau*ꜜ	*arau*		*arau_itigo*	"strawb to wash" (wash + strawb)	
(e)	*akaI*ꜜ	*akaI*		*akaI_itigo*	"red strawb" (red + strawb)	

To account for the absence of downsteps at the junctures in (13cde), Martin (1988: 22) derives phrasal patterns from PRs—instead of URs, as assumed in the standard analysis—of the building t-words of a phrase. In doing so, however, Martin's reanalysis predicts wrong results such as ˣ⟦*sasimi_arau*⟧, without a downstep, for cases like (13a).[15]

Similarly, ⟨=*hodo*⟩-type particles also pose a problem for Martin's reanalysis.

(14) Final-[❶] base + ⟨=*hodo*⟩ in Martin's analysis (problematic cases)

	[...=*hodo*] "as much as"	final-[❶] base			
		Martin's UR	gloss	last element	
(a)	❶❶❶=❶❶	*itigo*°	"strawb"	unaccented noun	
(b)	❶❶❶=❶❶	*arau*ꜜ	"wash"	verb	base-final downstep removed
(c)	❶❶❶=❶❶	*akaI*ꜜ	"be red"	*i*-adjective	
(d)	❶❶❶=❶=❶❶	*itigo=ni*ꜜ	"strawb=DAT"	particle	
(e)	❶❶❶=ⓁⓁ	*sasimi*ꜜ	"sashimi"	final-accented noun	base-final downstep retained

[15] Martin (1987: 150; 1988: 22) gives wrong ⟦*kagami_mi*ꜜ*te*⟧ instead of the correct ⟦*kagami*ꜜ*_mi*ꜜ*te*⟧ "Look at the mirror!" (mirror + look) in his illustration of the t-phrasing pattern using a bare-argumental t-phrase.

Martin (1988: 916, 1003 fn. 10) gives a far-fetched explanation: there is an underlying t-phrase boundary later cancelled at the juncture, which helps to remove the downstep from (14bcd)—but it doesn't apply to cases like (14e), when the base is a nominal. This is again stipulated rather than well-motivated.

2.5 Some other blends: McCawley (1965) and Okuda (1980)

There are also some theories that are logical (rather than genealogical) blends of the standard analysis and Martin's reanalysis, notably McCawley (1965) and Okuda (1980). They assume, like Martin's reanalysis, that final-[❶] particles are followed by downsteps; what makes them different from Martin's is that they assume certain final-[❶] inflectional forms of verbs and *i*-adjectives (for McCawley, all forms; for Okuda, only forms that are not further decomposable into a stem and a particle—including lemmas) are not followed by downsteps.

These varieties present both problems mentioned so far, instead of solving either of them. The reader may easily verify this by checking against (15).

(15) Final-[❶] base + ⟨=*hodo*⟩ and ⟨=*nara*⟩ in McCawley's and Okuda's analyses (problematic cases)

	[...=*hodo*] "as much as"	[...=*nara*] "if ..."	final-[❶] base		
			McCawley's & Okuda's UR	gloss	last element
(a)	❶❶❶=❶	❶❶❶=❶Ⓛ	*itigo*°	"strawb"	unaccented noun
(b)	❶❶❶=❶	❶❶❶=ⓁⓁ	*arau*°	"wash"	verb
(c)	❶❶❶=❶	❶❶▼=ⓁⓁ → ❶❶▽=ⓁⓁ	*akaI*°	"be red"	*i*-adjective
(d)	❶❶❶=❶=❶	❶❶❶=❶=ⓁⓁ	*itigo=ni*ꜜ	"strawb=DAT"	particle
(e)	❶❶❶=Ⓛ	❶❶❶=ⓁⓁ	*sasimi*ꜜ	"sashimi"	final-accented noun

2.6 Comparison of analyses

In this section I have reviewed existing analyses against the two major recessive t-cliticization patterns built on bases ending in various PoSs, finding none of them satisfactory. Obviously, final-[❶] bases' 3-way conditioning on recessive particle tonology, when forced into a 2-way distinction of [...°] vs. [...ꜜ] in the UR, is doomed to create a situation of robbing Peter to pay Paul. (16) is a summary.

(16) Final-[❶] bases' URs and conditioning on particle tonology

particle SR		final-[❶] base	base UR		
⟨=*hodo*⟩	⟨=*nara*⟩		standard	McCawley, Okuda	Martin
=❶❶	=❶Ⓛ	unaccented nominal	...°	...°	...°
=❶❶	=ⓁⓁ	verb/*i*-adjective	...°	...°	...ꜜ
=❶❶	=ⓁⓁ	particle	...°	...ꜜ	...ꜜ
=ⓁⓁ	=ⓁⓁ	final-accented nominal	...ꜜ	...ꜜ	...ꜜ

2.7 Adverbial usage effect

There is another catch for all existing theories: the adverbial usage effect of nouns. Consider the following tonal alternations in (17) with two interpretations of the bare noun:

(17) Adverbial usage effect (🅐: nondefective 🅗; problematic cases for Martin's reanalysis)

			final-nondef.	final-defective
			(i) ⟨*natu'*⟩ "summer"	(ii) ⟨*kino'O*⟩ "yesterday"
(a)	⟦_⟧	(citation form)	🅗🅐	🅗🅗▽
(b)	⟦_ *eraNda*⟧	(argumental usage) "(Someone) chose … ."	🅗🅐‿Ⓛ…	🅗🅗▽‿Ⓛ…
(c)	⟦_ *eraNda*⟧	(adverbial usage) "(Someone) chose in … ."	🅗🅐‿🅗…	🅗🅗▼‿Ⓗ…

In the standard analysis, this effect is captured by a "final syllable deaccenting" rule (Hirayama 1960: FM13; Akinaga 1966; 2014; NHK 2016). This is, again, arbitrary in nature—at least in the sense that it doesn't correspond to any surface-true generalizations about Tokyo or cross-linguistic tendencies.

Martin's reanalysis is even more problematic in this regard. His reanalysis doesn't acknowledge a distinct mechanism, but attributes the adverbial usage effect along with all other phrasal patterns to a simple concatenation of their PRs (with some form of a "final syllable deaccenting" rule implied therein). In doing so, theoretically Martin would only predict ⟦…🅐‿🅗…⟧ (= (17ci), for final-nondefective nouns) and ⟦…▽‿Ⓛ…⟧ (= (17bii), for final-defective nouns) for both adverbial and argumental usages alike, leaving ⟦…🅐‿Ⓛ…⟧ (17cii) and ⟦…▼‿Ⓗ…⟧ (17bi) unaccounted for. Worse still, Martin (1988: 21–22) actually lists usages parallel to (17bii) as ⟦…▽‿Ⓛ…⟧, contradicting his own formulation.

In addition, many descriptions of the effect in existing analyses (including Akinaga 1966; Martin 1988) are imprecise in certain aspects. For a fuller description and analysis of the adverbial usage effect, see § 3.5.2 and § 4.2.

3 Tokyo postlexical tonology in floating tone reanalysis

In this section, I will show that a floating tone reanalysis can elegantly account for the paradoxical pattern of Tokyo recessive t-cliticization. Before going into the reanalysis, I shall discuss why this reanalysis is not an arbitrary stipulation, but rather naturally follows from a generalization about recessive t-cliticization.

3.1 Nominal Overriding Generalization

From the failed analyses addressing the pattern shown in (16), one might conclude prematurely that a phonologically concrete analysis of Tokyo recessive t-cliticization is impossible. A closer look at the pattern, simplified as (18) below, suggests otherwise:

(18) Realization of the first mora of the particle for the 2 recessive particle types

			recessive particle type		
			⟨=*hodo*⟩-type	⟨=*nara*⟩-type	
(a)	final-[❶] base	unaccented nominal	❶	❶	contributed by the base
(b)		non-nominal	❶	Ⓛ	contributed by the particle
(c)		accented nominal	Ⓛ	Ⓛ	contributed by the base

From (18), the following generalization about Tokyo recessive t-cliticization can be stated:[16]

- **Nominal Overriding Generalization**: In recessive t-cliticization of Tokyo, unaccented nominal bases force a [❶] onto the first mora of the recessive particle, and accented nominal bases force an [Ⓛ].

From this generalization, the following analysis is inevitable:

- The [❶]s in (18a), and the [Ⓛ]s in (18c) come from the nominal base;
- The [❶] and the [Ⓛ] in (18b) come from the particle.

This can be translated into my reanalysis of Tokyo recessive t-cliticization, which incorporates the following two aspects:

- Representation-wise, only nominal bases systematically require a trailing floating tone at clitic input.
- Realization-wise, tone conflict is resolved in favour of the floating tone in recessive t-cliticization.

These two aspects are unpacked below in turn.

3.2 Floating tone reanalysis

3.2.1 Autosegmental representations

Recessive particles fall into two classes according to what tone is underlyingly specified on their first moras: ⟨=*hodo*⟩-type (initial-/❶/) and ⟨=*nara*⟩-type (initial-/Ⓛ/). Their URs are exemplified in (19).

(19) URs of recessive particles ⟨=*hodo*⟩ and ⟨=*nara*⟩

=❶ ❶	=Ⓛ Ⓛ
=↕ ↕	=↕ ↕
=*hodo*	=*nara*

Representations of lexical bases are exemplified in (20):

[16] Note that the Nominal Overriding Generalization is limited to recessive t-cliticization and doesn't apply to Tokyo tonology across the board. For example, the surname ⟨*matumoto*°⟩ doesn't force a ❶ on the first mora of ⟨-*si*⟩ "Mr/Ms …", a "preaccented" recessive derivational suffix: ⟨*matumoto*'-*si*⟩. (Hirayama 1960: 917; Akinaga 2014: #94)

(20) URs of lexical bases (↕: regular mora skeleton; ↟: floating skeleton)

	unaccented	accented		
		final-	medial-	initial-
noun	❶❶❶❶ ↕ ↕ ↕ ↟ *i ti go*	❶❶❶Ⓛ ↕ ↕ ↕ ↟ *sa si mi*	❶ ❶ ⓁⓁ ↕ ↕ ↕ ↟ *ta mago*	❶ Ⓛ ⓁⓁ ↕ ↕ ↕ ↟ *banana*
na-adjective (lemma)	❶❶❶❶ ↕ ↕ ↕ ↟ *hu si gi*	❶ ❶❶Ⓛ ↕ ↕ ↕ ↟ *ma Q ka*	❶ ❶ ⓁⓁ ↕ ↕ ↕ ↟ *to ku i*	❶ Ⓛ ⓁⓁ ↕ ↕ ↕ ↟ *si zuka*
i-adjective (lemma)	❶❶▼ ↕ ↕ ↕ *a ka I*	❶❶Ⓛ ↕ ↕ ↕ *si ro I*		
verb (lemma)	❶❶❶ ↕ ↕ ↕ *a ra u*	❶❶Ⓛ ↕ ↕ ↕ *ta beru*		

In (19) and (20), ↕s are regular mora skeletons, whereas ↟s are floating skeletons. The floating skeleton is proposed to model the obligatory floating tone at the end of nominal bases. Floating skeletons are similar to regular skeletons in requiring tone specification, but lack segmental specification completely. When the base enters into recessive t-cliticization, the floating skeleton is overlaid on the corresponding regular mora skeleton of the particle, causing overcrowding of tones and triggering tone relinking.

3.2.2 Recessive t-cliticization

The morphophonological processes involved in t-cliticization consist of two stages: morphological composition and phonological repair.

The morphological composition process of recessive t-cliticization in turn involves two steps:

- Skeleton assembly (recessive): Load the base, including the segments and the tones, before the particle. Concatenate regular skeletons, and let floating skeletons align accordingly.
- Tone relinking (floating-win[17]): In the case of tone conflict caused by the floating tone, link the mora to the floating tone of the base and delink the mora from the other tone.

For cases without a floating tone, this process is trivial; for cases with a floating tone, the composition process is illustrated in (21).

(21) Morphological composition of ⟨*itigo=nara*⟩: recessive assembly and floating-win relinking

	base		particle		recessive assembly		floating-win relinking		result	
(a)	❶❶❶❶ ↕ ↕ ↕ ↟ *i ti go*	+	=Ⓛ Ⓛ = ↕ ↕ *=nara*	=	❶❶❶=❶Ⓛ Ⓛ ↕ ↕ ↕= ↕ ↕ *i ti go= na ra*	→	❶❶❶=❶⃠Ⓛ ↕ ↕ ↕= ↕ ↕ *i ti go= na ra*	=	❶❶❶=❶Ⓛ ↕ ↕ ↕= ↕ ↕ *i ti go=nara*	diagram notation
(b)	❶❶❶◖	+	=Ⓛ Ⓛ	=	❶❶❶◖Ⓛ Ⓛ	→	❶❶❶⃠Ⓛ	=	❶❶❶=❶Ⓛ	inline notation

For ⟨*itigo=nara*⟩:

- Recessive skeleton assembly (❶❶❶◖Ⓛ Ⓛ): The resulting composite has 5 moras rather than 6, because the 4th mora of the noun base, as a floating skeleton ↟, is overlaid on the first mora of the particle, and

[17] Which can be construed as either left-win or base-win.

doesn't contribute to the eventual mora count.

- Floating-win tone relinking (❶❶❶◖Ⓛ): The 4th mora, segmentally linked to *na*, is tonally relinked to the floating tone ◖ supplied by the base, with the original docked tone Ⓛ of the particle deleted.

For convenience' sake, diagram notations like (21a) will henceforth be given alternative inline notations as shown in (21b). Tone clashes after skeleton assembly and tone relinking are iconically represented in inline notation as ◖Ⓛ and ◖◹ etc.[18], and floating tones are accordingly represented as ◖ etc. in inline notation even when standing alone.

After that, phonological repair is responsible for transforming the morphologically composed representations into well-formed PRs:

(22) Repair rules in cliticization

- **Monotonic Lowering**: ❶ → Ⓛ / Ⓛ_ (Any pitch level after a low pitch level changes to low. Iterative.)
- Defective Lowering: ▼ → ▽ / _Ⓛ (A high-toned defective mora is lowered before a low tone. Iterative.[19])

Monotonic Lowering and Defective Lowering ensure that the Monotonicity Constraint and the Defectiveness Constraint (see § 1.1) are satisfied.

Recessive patterns built on bases of various PoSs are given as (23)~(24) below. For each combination, the phonological derivation process is listed from top to bottom in the cell.

(23) Derivation of recessive patterns built on verb & *i*-adjective lemma bases

	verb lemma, unaccented	*i*-adj. lemma, unaccented	verb lemma, accented	*i*-adj. lemma, accented
	❶❶❶ ↕ ↕ ↕ *a ra u*	❶❶▼ ↕ ↕ ↕ *a ka I*	❶❶Ⓛ ↕ ↕ ↕ *taberu*	❶❶Ⓛ ↕ ↕ ↕ *si ro I*
=❶❶ =↕ ↕ =*hodo*	❶❶❶=❶❶	❶❶▼=❶❶	❶❶Ⓛ=❶❶ ❶❶Ⓛ=ⓁⓁ	❶❶Ⓛ=❶❶ ❶❶Ⓛ=ⓁⓁ
=ⓁⓁ =↕ ↕ =*nara*	❶❶❶=ⓁⓁ	❶❶▼=ⓁⓁ ❶❶▽=ⓁⓁ	❶❶Ⓛ=ⓁⓁ	❶❶Ⓛ=ⓁⓁ

↝ Monotonic Lowering

↝ Defective Lowering

Recessive patterns built on verb and *i*-adjective bases are the most straightforward: there is no floating tone and thus no relinking.

- /*taberu=hodo* ❶❶Ⓛ=❶❶/ → [❶❶Ⓛ=ⓁⓁ]: The high tones of *hodo* are lowered after low-toned *ru* through Monotonic Lowering.

(24) Derivation of recessive patterns built on nominal bases

	unaccented	accented		
		final-	medial-	initial-
	❶❶❶❶ ↕ ↕ ↕ ↟ *i ti go*	❶❶❶Ⓛ ↕ ↕ ↕ ↟ *sa si mi*	❶ ❶ ⓁⓁ ↕ ↕ ↕ ↟ *tamago*	❶ Ⓛ ⓁⓁ ↕ ↕ ↕ ↟ *banana*

[18] With the boundary marker (=) omitted. In this and similar cases, segmentally the boundary is before the eclipsed symbol, and tonally the boundary is after it.

[19] Iterative as in ⟨*toOI=nara*⟩ (far=if) /❶▼▼=ⓁⓁ/ → [❶▽▽=ⓁⓁ].

=Ⓗ Ⓗ = ↕ ↕ =*hodo*	ⓗⓗⓗ◖ⓗ ⓗⓗⓗ=ⓗⓗ	ⓗⓗⓗ◖ⓗ ⓗⓗⓗ=Ⓛⓗ ⓗⓗⓗ=ⓁⓁ	ⓗⓗⓁ◖ⓗ ⓗⓗⓁ=Ⓛⓗ ⓗⓗⓁ=ⓁⓁ	ⓗⓁⓁ◖ⓗ ⓗⓁⓁ=Ⓛⓗ ⓗⓁⓁ=ⓁⓁ	⟆ relinking ⟆ Monotonic Lowering
=Ⓛ Ⓛ = ↕ ↕ =*nara*	ⓗⓗⓗ◖Ⓛ ⓗⓗⓗ=ⓗⓁ	ⓗⓗⓗ◖Ⓛ ⓗⓗⓗ=ⓁⓁ	ⓗⓗⓁ◖Ⓛ ⓗⓗⓁ=ⓁⓁ	ⓗⓁⓁ◖Ⓛ ⓗⓁⓁ=ⓁⓁ	⟆ relinking

For nominal bases, there is a trailing floating skeleton at the end of the base, causing relinking of the tones: both ◖ and ◖ resolve to Ⓗ; both ◖ and ◖ resolve to Ⓛ. After relinking:

- */sasimi=hodo* ⓗⓗⓗ=ⓁⒽ/ → [ⓗⓗⓗ=ⓁⓁ]: The high-toned *do* is lowered after the low-toned *ho*, through Monotonic Lowering.

3.2.3 T-phrasing

The floating tone reanalysis, like the standard analysis, allows us to derive phrasal patterns elegantly as shown in (25):

- **T-Phrasing** (reformulated as a rule): Continue the first pitch level of a t-word at the last pitch level (floating tones included) of the preceding t-word according to their SRs.

(25) T-phrasing in floating tone reanalysis (Ⓗ: Initial Lowering)

	visualized pitch	SR	gloss
(a)	Ⓗ ⓗⓗ ◖ⓗⓗ Ⓛ	ⓗⓗⓗ◖ + ⓗⓗⓁ *sasimi* + *taberu*	“eat sashimi” (sashimi + eat)
(b)	Ⓗ ⓗⓗ◖ⓗⓗ Ⓛ	ⓗⓗⓗ◖ + ⓗⓗⓁ *itigo* + *taberu*	“eat strawb” (strawb + eat)
(c)	Ⓗ ⓗⓗ=Ⓗ‿ⓗⓗ Ⓛ	ⓗⓗⓗ=Ⓗ + ⓗⓗⓁ *itigo=o* + *taberu*	“eat strawb” (strawb=ACC + eat)
(d)	Ⓗ ⓗ Ⓛ‿ⓗⓗ Ⓛ	ⓗⓗⓁ + ⓗⓗⓁ *taberu* + *tamago*	“egg to eat” (eat + egg)
(e)	Ⓗ ⓗ Ⓛ‿ⓗⓗ Ⓛ	ⓗⓗⓁ + ⓗⓗⓁ *siroI* + *tamago*	“white egg” (white + egg)

3.3 Apparent exceptions to the Nominal Overriding Generalization

As the starting point of my reanalysis, the Nominal Overriding Generalization has, however, eluded all previous theories to the best of my knowledge: most of them formally allow certain potential recessive particles (usually called “preaccented” or “preaccenting” particles) to surface as invariably initial-[Ⓛ], and many of them (including Akinaga 1966: 58; McCawley 1977: 263) actually misanalyse some as such. These are discussed and dismissed below in turn.

⟨=***sika***⟩ (“except …”). This particle is underlyingly /=ⓁⓁ/, surfacing as [=ⒽⓁ] after an unaccented noun. However, according to some (including Akinaga 2014), its surface form should be identified as [=ⓁⓁ]. As noted in Kori (2020), this is actually a misinterpretation of its phonetic realization, ⟦=◍Ⓛ⟧, because the first mora of the particle *si̥* is typically devoiced. Crucial evidence comes from Vance (1987: 84), where he notes that ⟨*gaQkoO=si̥ka*⟩ “except school” is phonetically ⟦ⓗ◍ⓗⒽ=◍Ⓛ⟧, which must come from [ⓗⓗⓗⒽ=ⒽⓁ] at a deeper level. If it were underlyingly ˣ[ⓗⓗⓗⒽ=ⓁⓁ], through Defective Lowering (and vowel devoicing) it would change to ˣ⟦ⓗ◍ⓗⓁ=◍Ⓛ⟧, which is not the case.

⟨=***to***⟩ (the quotation marker). Inferring from its realizations in indirect quotation and related[20] usages as listed in (26bii), we know this particle is underlyingly /=❶ ~ =Ⓛ/. However, as Kori (2020) and Akinaga (2014: #71) note, after a directly quoted unaccented noun, in addition to the regular [=❶] realization ([*itigo=to* ❶❶❶=❶] " 'strawberry' "), [=Ⓛ] is exceptionally allowed ([*itigo=to* ❶❶❶=Ⓛ]), as in (26ai).

(26) Realizations of /=*to* =❶ ~ =Ⓛ/ in direct quotation and other usages

			(i) direct q.	(ii) elsewhere
(a)	final-[❶] base	unaccented nominal	❶~Ⓛ	❶
(b)		non-nominal	❶~Ⓛ	❶~Ⓛ
(c)		accented nominal	Ⓛ	Ⓛ
(d)	final-[Ⓛ] base		Ⓛ	Ⓛ

To account for this exception, note the generalization that only nominals possess a trailing floating skeleton. Direct quotation, being **mention** rather than **use**, is indifferent to the internal PoS label of the quotation. Therefore, it is not unimaginable for the quoted noun to optionally drop the floating skeleton before being integrated into the context. With /*itigo* ❶❶❶◖/ dropping the trailing floating skeleton, now the exceptional /"*itigo* ❶❶❶"/ + /=*to* =Ⓛ/ → [*itigo=to* ❶❶❶=Ⓛ] is obtained.

3.4 URs of recessive particles

The URs of particles are listed in (27). Normally they can be recovered by directly examining their surface tones after an unaccented (i.e., final-[❶]) verb or *i*-adjective, because there's neither a floating tone from the base to trigger relinking, nor a low tone in the base to lower succeeding tones through Monotonic Lowering. There are a few exceptions, though, for which see dedicated discussions below. For ⟨=*no*⟩, see § 3.6.

(27) URs of recessive particles[21]

type	UR	list
⟨=*nara*⟩	=Ⓛ(Ⓛ...)	(the majority)
	=Ⓛ (with irregularity)	=^{na}*no*
	=Ⓛ❶	=^{da}*kara*
⟨=*nara*⟩ ~ ⟨=*hodo*⟩	=Ⓛ❶Ⓛ (innovative) ~ =❶❶Ⓛ (conservative)	=^{0}*daroO* =^{0}*desyoO*
	=Ⓛ(Ⓛ) ~ =❶(Ⓛ)	=*sae* =*made* =^{na}*yori* =$^{(da)}$*to* (=QUOT) =$^{(da)}$*yo* =^{da}*ze* =^{da}*zo*
⟨=*hodo*⟩	=❶(Ⓛ...)	=^{na}*nomi* =^{na}*bakari* =$^{no/na}$*yoO*^{na} =^{0}*mitai*^{na} =^{da}*soO*^{na} =^{da}*na* =$^{(da)}$*ne* =*gotoni* =*gotono*
	=❶(Ⓛ...) (also dominant)	=^{0}*dokoroka* =^{0}*dokorono* =^{0}*rasiku* =^{na}*kuraI/guraI*

[20] Complementization, enumeration, and composite particles derived therefrom like ⟨=*to=ka*⟩ "...or the like" (lit. =enumeration=interrogation).

[21] Prefixal ^{na}, ^{da}, ^{0}, etc. denote the base forms of *na*-adjectives and the copular selected by the particle.

=❶❶(Ⓛ) (also dominant)	=^{0}*rasiI* =^{na}*dake*
=❶❶ ~ =❶❶◖	=^{na}*hodo*

⟨=***hodo***⟩: /=❶❶ ~ =❶❶◖/. Kori (2020) notes that ⟨=*hodo*⟩ can optionally inhibit a downstep at its right edge when followed by another particle (even when that particle belongs to ⟨=*nara*⟩-type). This means ⟨=*hodo*⟩ is not yet a fully grammaticalized particle and still preserves some residue of its original noun property. This characteristic is captured in my reanalysis with an optional trailing floating ◖ (though not supported by a floating skeleton), i.e. /=*hodo* =❶❶ ~ =❶❶◖/.

⟨=***kara***⟩: /=Ⓛ❶/. ⟨=*kara*⟩ has two etymologically related functions: as the ablative particle after nouns, and as the conjunctive particle marking causes (after the indicative finite) or previous events (after the *te*-form of verbs). ⟨=*kara*⟩ with these two functions, if identified as a single item in synchronic analysis, is the only one in conservative Tokyo[22] of ⟨=*nara*⟩-type surfacing as initial-[❶❶] after unaccented nouns, in contrast to all others surfacing as initial-[❶(Ⓛ…)] in the same context (see (28)). This special conditioning of ⟨=*kara*⟩ can actually be derived from a single UR, /=Ⓛ❶/, although this underlying sequence Ⓛ❶ is never realized as such in surface forms.

(28) Comparison of realizations of ⟨=*nara*⟩ and ⟨=*kara*⟩

		⟨=*nara*⟩	⟨=*kara*⟩	
final-[❶] base	unaccented nominal	=❶Ⓛ	=❶❶	← ◖⃠❶
	non-nominal	=ⓁⓁ	=ⓁⓁ	← =Ⓛ❶
	accented nominal	=ⓁⓁ	=ⓁⓁ	← =Ⓛ❶ ← ◖⃠❶
final-[Ⓛ] base	accented nominal	=ⓁⓁ	=ⓁⓁ	← =Ⓛ❶ ← ◖⃠❶
	non-nominal	=ⓁⓁ	=ⓁⓁ	← =Ⓛ❶

⟨=***daroO***⟩, ⟨=***desyoO***⟩: /=❶❶Ⓛ/ (conservative) ~ /=Ⓛ❶Ⓛ/ (innovative). ⟨=*daroO*⟩ is the suppositional form of the copula, whereas ⟨=*desyoO*⟩ is its polite counterpart, tonologically perfectly parallel to it. ⟨=*daroO*⟩ has two surface forms, [=❶❶Ⓛ] and [=ⓁⓁⓁ], whose distribution varies across conservative and innovative variants, as listed in (29). In conservative Tokyo, their distributions parallel those of ⟨=*hodo*⟩; in innovative Tokyo, their distributions parallel those of ⟨=*kara*⟩. (NHK 2016: app. 221; Kori 2020) Therefore, the UR should be analysed as /=❶❶Ⓛ/ in the conservative variant and /=Ⓛ❶Ⓛ/ in the innovative variant.

(29) Comparison of realizations of ⟨=*hodo*⟩, ⟨=*kara*⟩, ⟨=*daroO*⟩

		⟨=*daroO*⟩ (conserv.)	⟨=*hodo*⟩	⟨=*kara*⟩	⟨=*daroO*⟩ (innov.)	
final-[❶] base	unaccented nominal	=❶❶Ⓛ	=❶Ⓛ	=❶❶	=❶❶Ⓛ	← ◖⃠❶Ⓛ
	non-nominal	=❶❶Ⓛ	=❶Ⓛ	=ⓁⓁ	=ⓁⓁⓁ	← =Ⓛ❶Ⓛ
	accented nominal	=ⓁⓁⓁ	=ⓁⓁ	=ⓁⓁ	=ⓁⓁⓁ	← =Ⓛ❶Ⓛ ← ◖⃠❶Ⓛ
final-[Ⓛ] base	nominal	=ⓁⓁⓁ	=ⓁⓁ	=ⓁⓁ	=ⓁⓁⓁ	← =Ⓛ❶Ⓛ ← ◖⃠❶Ⓛ
	non-nominal	=ⓁⓁⓁ	=ⓁⓁ	=ⓁⓁ	=ⓁⓁⓁ	← =Ⓛ❶Ⓛ

This change from /=❶❶Ⓛ/ to /=Ⓛ❶Ⓛ/ was probably driven by the fact that all other forms of the copula have a UR beginning with /Ⓛ/ (Martin 1952: 45–46): /=*da* =Ⓛ/ (lemma), /=*de* =Ⓛ/ (*te*-form), /=*daQta* =ⓁⓁⓁ/ (past), /=*na* =Ⓛ/ (adnominal), /=*nara* =ⓁⓁ/ (hypothetical), /=*desu* =ⓁⓁ/ (polite).

[22] In innovative Tokyo it is paralleled by ⟨=*daroO*⟩ and ⟨=*desyoO*⟩, for which see below.

3.5 Adverbs and adverbial usage of nouns

3.5.1 Adverbs

So far we have addressed all major lexical PoSs of Japanese except adnominals and adverbs. Adnominals do not admit particle cliticization and are thus uninteresting for our discussion due to a lack of morphophonological alternation.

Typical adverbs don't force a tone on a following particle (30a), which means they lack an underlying trailing floating tone.

(30) Adverb + /=*desu* =ⓁⓁ/ (copula) (Akinaga 1966: 81; Akinaga 2014: #76, #86)

	base UR	[…=*desu*]	example	
(a)	❶❶❶❶	❶❶❶❶=ⓁⓁ	*maQtaku* *seQkaku*	"completely" "at great pains"
(b)	❶❶❶❶ ~ ❶❶❶❶◖	❶❶❶❶=ⓁⓁ ❶❶❶❶=❶Ⓛ	*tokidoki*	"sometimes"

There are also exceptions that exhibit vacillation (like (30b)). These exceptional cases can be viewed as incomplete conversions from nouns, thus still preserving the trailing floating high tone ◖ despite the lack of a supporting floating skeleton.[23]

3.5.2 Adverbial usage effect and Bare Adverbial Defloatingization

The adverbial usage effect, which I have touched upon lightly in § 2.7, appears to be related to the floatingless nature of adverbs. Recall that the effect poses a problem for all existing analyses. I now show how the floating tone reanalysis captures it naturally.

A wider survey of nouns having bare adverbial usage, classified according to usages (adverbial vs. argumental), phonological context (phrased vs. isolated vs. cliticized), and UR shapes (final-defective vs. final-nondefective), yields the following pattern shown in (31):

(31) Adverbial usage effect on variously shaped URs in various phonological contexts

				kinoO "yesterday"	*natu* "summer"	*baN* "evening"	*yoru* "evening"
				/…▼◖/	/…🔺◖/	/…❶◖/	/…Ⓛ◖/
(a)	⟦… + *eraNda*⟧ (… + chose)	phrased	adv. usage	…▼‿❶…	…🔺‿❶…	…❶‿❶…	…Ⓛ‿Ⓛ…
			arg. usage	…▽‿Ⓛ…	…🔺‿Ⓛ…		
(b)	⟦…⟧ (…)	isolated	adv. usage	…▼	…🔺	…❶	…Ⓛ
			arg. usage	…▽			
(c)	⟦…=*mo*⟧ (…=even)	cliticized	adv. usage	…▽‿Ⓛ	…🔺‿Ⓛ	…❶‿❶	…Ⓛ‿Ⓛ
			arg. usage				

As shown above, adverbial usage non-vacuously affects:

- phrased and isolated (31ab) nouns with a UR of /…▼◖/ (with a defective final mora),
- only phrased (31a) nouns with a UR of /…🔺◖/ (with a nondefective final mora), and
- no other words.

[23] Rather than forming a well-delineated class, adverbs have a somewhat blurred border with nouns. This is because (i) adverbs are a relatively open class with new members constantly formed from bare nouns and particle-ending phrases, and (ii) some nouns have ready adverbial usage.

This adverbial usage effect has been formulated as a "final syllable deaccenting" rule in previous analyses, notably Hirayama (1960), Akinaga (1966; 2014), Martin (1988), and NHK (2016). It should be noted, however, that many of these formulations are imprecise, especially regarding the following two points:

- Phonologically isolated nouns are not exempt from the adverbial usage effect (pace Martin 1988: 21). As the example in Matsumori et al. (2012: 50, ex. 8ab) shows, when /*ototoI* ❶❶❶▼◟/ "ereyesterday" is used adverbially and uncliticized, it surfaces as ⟦❶❶❶▼⟧ because of the adverbial usage effect (instead of as ⟦❶❶❶▽⟧, its citation form), whether it is phonologically phrased with a following independent word.
- Nouns followed by particles are exempt from the adverbial usage effect, as noted in Hirayama (1960: FM13). This condition has eluded many past formulations of the effect as a lexical deaccenting rule (including Akinaga 2014: #62; NHK 2016: app. p. 84).

The adverbial usage effect can be captured in my floating tone reanalysis as the following rule:

- **Bare Adverbial Defloatingization** (BAD): When a t-word is used adverbially (rather than as an argument), delete its trailing floating skeleton along with its tone.

Compared with previous formulations of the adverbial usage effect as an arbitrary final syllable deaccenting rule, this new formulation aptly contextualizes the effect within the floatingless nature of adverbs and conspires with the well-formedness constraint, thus having higher explanatory power.

A further question is: which nouns are affected non-vacuously by this rule? We only know from (31) that all that are affected by BAD are accented on the last syllable (henceforth **last-σ-accented**). The reverse, however, is not true, because theoretically nouns with an underlying ▽ (defective low tone) before the floating tone (as in (32c)), though last-σ-accented, will not be affected by BAD according to the exact formulation of the rule given above:

(32) Classification of last-σ-accented nouns by whether they are theoretically affected by BAD

		PR of arg. usage	PR of adv. usage	UR	
(a)	final-nondefective	…❹‿Ⓛ…	…❹‿❶…	…❹◟	theoretically affected by BAD
(b)	final-defective	…❹▽‿Ⓛ…	…❹▼‿❶…	…❹▼◟	
(c)		…❹▽‿Ⓛ…	…❹▽‿Ⓛ…	…❹▽◟	theoretically unaffected by BAD

Unfortunately, the underlying distinction between (32b) and (32c) doesn't manifest itself elsewhere in postlexical Tokyo tonology. This makes a final-defective last-σ-accented noun underdetermined with respect to the UR (and in turn, underdetermined with respect to whether it is affected by BAD) a priori. A survey of last-σ-accented quantitative and temporal nouns—the two biggest groups of nouns systematically having bare adverbial usage—shows that, among final-defective last-σ-accented nouns:

- Polysyllabic ones (33b) and ⟨*toO*⟩ (33c) are affected by BAD, and
- Monosyllabic ones except ⟨*toO*⟩ (33d) are unaffected by BAD. (Hirayama 1960: FM13)

We may therefore deduce, among final-defective last-σ-accented nouns:

- Polysyllabic ones (33b) are underlyingly /…❹▼◟/,
- ⟨*toO*⟩[24] (33c) is underlyingly /❹▼◟/, and

[24] ⟨*toO*⟩ was pronounced as ⟨*to.o*⟩, ending in a disyllabic sequence instead of in a defective mora, till the end of Late Middle Japanese. (Frellesvig 2010: 323) Therefore, the distinction between /…❹▼◟/ and /❹▽◟/ (and in turn, the "affectedness condition of BAD") was once predicable from the syllable count (polysyllabic → /…❹▼◟/, affected by BAD; monosyllabic → /❹▽◟/, unaffected by BAD), although today, after the ⟨*to.o*⟩ → ⟨*toO*⟩ sound change, the distinction has become fully lexical and needs to be coded underlyingly.

- Monosyllabic ones except ⟨*toO*⟩ (33d) are underlyingly /❹▽◖/.

(33) Examples of last-σ-accented quantitative and temporal nouns

		(i) quantitative		(ii) temporal		
(a)	final-nondefective, polysyllabic (⇒ final-accented)	*zyuUiti'*	"11"	*asita'*	"tomorrow"	/...❹▼◖/, affected by BAD
		hyaku'	"100"	*yuUbe'*	"last night"	
		hutari'	"2 people"	*natu'*	"summer"	
		itido'	"once"	*huyu'*	"winter"	
(b)	final-defective, polysyllabic (⇒ medial-accented)	*goju'U*	"50"	*ototo'I*	"ereyesterday"	
		saNni'N	"3 people"	*kino'O*	"yesterday"	
		takusa'N	"a lot"			
		oOze'I	"a lot of people"			
(c)	final-defective, monosyllabic (⇒ initial-accented)	*to'O*	"10 pieces"	—	—	/❹▽◖/, unaffected by BAD
(d)		*yo'N*	"4"	*kyo'O*	"today"	
		kyu'U	"9"			
		zyu'U	"10"			
		se'N	"1000"			

BAD has a special status in the architecture of Tokyo tonology. It is not a phrasal domain rule, because it bleeds T-Phrasing; it is not a lexical rule either, because it is bled by cliticization. It is a non-cliticization rule belonging to the clitic group domain.

3.6 *no*-deaccenting

Among all recessive particles, ⟨=*no*⟩ is a rather exceptional one that deserves special discussion. It serves a series of interrelated functions: the genitive marker, the adnominal form of the copula, the empty noun (nominalizer), and a final particle. By default, it behaves like other recessive particles that are underlyingly /=Ⓛ/.

However, complications arise when ⟨=*no*⟩ is attached to polysyllabic last-σ-accented nouns. In this case, there are two potential results: either it deaccents the base and renders the t-word unaccented (as in (34a)), or it behaves just like other /=Ⓛ/ recessive particles (as in (34b)).

(34) Polysyllabic last-σ-accented noun + ⟨=*no*⟩: idiosyncratically deaccenting vs. regularly recessive

		final-accented (final-nondefective)			medial-accented (final-defective)		
		[...=*no*]	base		[...=*no*]	base	
(a)	idiosyncratically deaccenting	❽❹=❽	*hyaku'*	"100"	❽❹▼=❽	*goju'U*	"50"
(b)	regularly recessive	❽❹=Ⓛ	*natu'*	"summer"	❽❹▽=Ⓛ	*kino'O*	"yesterday"

These two t-cliticization patterns of ⟨=*no*⟩ are selected by the base. The majority of polysyllabic last-σ-accented nouns follow the idiosyncratically deaccenting pattern. However, there are a few bases that select the regularly recessive pattern, including: (Akinaga 2014: #71)

- All quantitative nouns (cf. (33i)):
  - ⟨*hyaku'*⟩ "100", ⟨*goju'U*⟩ "50", ⟨*oOze'I*⟩ "a lot of people", ...;
- Certain historical proper nouns:
  - (old province names) ⟨*tosa'*⟩ "Tosa", ⟨*sinano'*⟩ "Shinano", ...
  - (stage names) ⟨*eNtyo'O*⟩ "Enchō", ⟨*baIko'O*⟩ "Baikō", ...;
- (optionally) Deverbal nouns from verbs ending in *w*:

    - ⟨*omoˈI*⟩ “thought” (from ⟨*omowᵛ*⟩), ⟨*negaˈI*⟩ “wish” (from ⟨*negawᵛ*⟩), …
- Some miscellaneous words:
    - ⟨*tugiˈ*⟩ “next”, ⟨*yosoˈ*⟩ “elsewhere”, ….

Unlike BAD, the lexical applicability of *no*-deaccenting cannot be coded in the bases’ URs—unless additional ad hoc phonological structures are posited. This is clear from the fact that ⟨*hyakuˈ*⟩ “100” and ⟨*natuˈ*⟩ “summer”, both being /❶❹◔/ underlyingly, have different tonal realizations when followed by ⟨=*no*⟩ (regularly recessive [*hyaku=no* ❶❶=Ⓛ] “of 100” vs. idiosyncratically deaccenting [*natu=no* ❶❶=❶] “of summer”). Given the complex semantic and lexical conditioning of *no*-deaccenting, we may dismiss *no*-deaccenting as idiosyncratic and having little bearing on the rest of Tokyo tonology.

3.7 The infinitives, the *te*-form, and the *akakù=wa* puzzle

All the bases addressed so far are lemmas. The next most frequent cliticization bases are the adverbial forms of inflectables. There are 3 adverbial forms, here termed the **free infinitive**, the **bound infinitive**, and the ***te*-form**. The *te*-form is the default form used, with or without particles, for continuing a sentence in Modern Japanese. Its competing forms, the free and bound infinitives, historically serving largely the same functions, remain today only in several restricted usages:

- The free infinitive, never followed by particles, competes with the *te*-form in literary Modern Japanese. In addition,
    - The free infinitive of an adjective serves to:
        - modify another predicate semantically like an adverb (i.e., without a clause structure), or
        - complement certain auxiliary predicates like ⟨… *narᵛ*⟩ “to become …”.
    - The free infinitive of a verb, however, is only residual in colloquial Tokyo, because it has no functions comparable to those of an adjective.[25]
- The bound infinitive, always followed by a focusing particle ⟨*wa/mo/sae/…*⟩ “alone/also/even/…”, only serves to complement the split construction ⟨…*=wa/mo/sae/… siᵛ/arᵛ*⟩.[26]

The three adverbial forms of verbs, *i*-adjectives, and *na*-adjectives are listed in (35) below:

(35) The 3 adverbial forms of unaccented inflectables (historical mutation) (Akinaga 2014: #72)

		(i) ⟨*arawᵛ*⟩ “to wash”		(ii) ⟨*akaⁱ*⟩ “to be red”		(iii) ⟨*himaⁿᵃ*⟩ “to be idle”	
(a)	free infinitive	*ara-i*	❶❶❶	*aka-ku*	❶❶-❶	*hima-ni*	❶❶-❶
(b)	b. inf. + ⟨=*wa*⟩	*ara-i=wa*	❶❶❶=Ⓛ	*aka-ku=wa*	❶❶-❶=Ⓛ ~ ❶❶-Ⓛ=Ⓛ	*hima-de=wa*	❶❶-❶=Ⓛ
(c)	*te*-form	*araQ-te*	❶❶❶-❶	*aka-ku-te*	❶❶-Ⓛ-Ⓛ	*hima-de*	❶❶-❶

The trickiest point here is that the bound infinitive doesn’t appear to have a “dedicated form” of its own, but is always syncretized either:

- to the free infinitive (*ara-i* ❶❶❶) (35ai), of a verb; or
- to the *te*-form (*hima-de* ❶❶-❶) (35ciii), of a *na*-adjective; or
- segmentally to the free infinitive (*aka-ku*) (35aii), but tonally to either the free infinitive or the *te*-form (❶❶-❶ or ❶❶-Ⓛ) (35a/cii), of an *i*-adjective.

[25] Verbs don’t modify another predicate semantically like an adverb; verbs use inceptive constructions ⟨… *ikᵛ*⟩ “to go to …” and ⟨… *koᵛ*⟩ “to come to …” based on the *te*-form.

[26] Japanese doesn’t readily allow non-adverbial forms of predicates to be followed by focusing particles. Therefore, whenever there’s a need to apply a focusing particle to a predicate that is not in an adverbial form, this split construction is substituted to provide an adverbial host for it. This is reminiscent of *do*-support in English—*says* : *does not say*.

Basing their analysis solely on segmental spelling and failing to note the tonal difference between [*akaku=wa* ❶❶Ⓛ=Ⓛ] and [*akaku* ❶❶❶], grammarians recognize only two separate adverbial forms of inflectables—the (free) infinitive and the *te*-form—and attribute the conditioned syncretism of the bound infinitive to some contextual syntactic rules external to the inflectional paradigm. Circularly misled by this analytical tradition, phonologists have long struggled to find a satisfactory account of the blended syncretism of [*akaku=wa* ❶❶Ⓛ=Ⓛ] (McCawley 1965: 157; Martin 1988: 22; Vance 1987: 97).

With hindsight, establishing the bound infinitive as a separate cell in the inflectional paradigm should hardly be surprising. First, the segmental split between the two infinitives of *na*-adjectives (*hima-ni* vs. *hima-de*=) independently justifies the morphosyntactic conditions behind the two cells. Second, both the *te*-form and the bound infinitive construction (like ⟨*akaku=wa*⟩) historically evolved from a concatenation of the (free) infinitive + particles, through a series of segmental and tonal mutations (coloured green in (35)). Therefore, if the *te*-form is to be recognized as an independent inflectional cell, I don't see why the bound infinitive deserves different treatment.

3.7.1 Monomoraic infinitives

The formation of the (free or bound) infinitive of verbs is listed below in (36):

(36) Infinitive of verbs (+ /=*wa* =Ⓛ/)

		unaccented	accented
		…❶❶(=Ⓛ)	…❶Ⓛ(=Ⓛ)
athematic	…*-i*	*nor-i*(=ˈ*wa*)	*kaˈk-i*(=*wa*)
thematic	…*-∅*	*abi*(=ˈ*wa*)	*taˈbe*(=*wa*)

For unaccented verbs, the accentuation is constantly […❶❶], and for accented verbs, normally […❶Ⓛ].

However, complications arise when an accented infinitive is monomoraic[27], such as ⟨*mi*⟩ "to see", the infinitive of ⟨*miᵛ*⟩. Obviously, the otherwise expected pattern of […❶Ⓛ] doesn't fit onto the monomoraic infinitive ⟨*mi*⟩ due to a lack of space. In that case, the actual accentuation of the infinitive is [❶◹], with a trailing ◹ capable of forcing a low tone on the following particle (37ii). In contrast, its unaccented counterpart ⟨*ne*⟩ "to sleep" doesn't have a trailing floating tone, leaving the first mora of the particle faithfully specified by its UR (37i).

(37) Monomoraic bound infinitive + particle (Akinaga 2014: #72)

		unaccented	accented
		(i) *ne* "to sleep" ❶	(ii) *mi* "to see" ❶◹
=*wa* (=TOP)	=Ⓛ	❶=Ⓛ	❶=Ⓛ
=*sae* "even"	=ⓁⓁ~ =❶Ⓛ	❶=ⓁⓁ~ ❶=❶Ⓛ	❶=ⓁⓁ

3.8 Putting together

The relative order of T-Cliticization, BAD, T-Phrasing, and then Floating Removal is quite clear and intuitive. While Monotonic Lowering can be subsumed under t-cliticization, the same cannot be assumed for Defective Lowering: as [*takusaN* ❶❶❶▼] (the adverbial usage of /*takusaN* ❶❶❶▼◹/ "a lot", affected by

[27] Only thematic verbs may have a monomoraic infinitive. In addition, monomoraic infinitives hardly ever stand alone due to the minimality constraint of predicate forms in Modern Japanese.

BAD) shows, Defective Lowering is bled by BAD, and thus needs to be ordered after it.

One possible order of all rules is listed in (38) below.

(38) Ordered list of all rules

- T-Cliticization (morphological composition, Monotonic Lowering)
- BAD
- Defective Lowering
- T-Phrasing
- Floating Removal

Although some other orders may be possible (e.g., the relative order between T-Phrasing and Defective Lowering is free), the choice among them cannot be settled without resorting to metatheoretical considerations.

The full derivation processes of ⟨*kinoˈO*⟩ and ⟨*asitaˈ*⟩ in various contexts are listed in (39)~(40) below:

(39) Full derivation of /*kinoO* ❶❶▼☾/ "yesterday" in various contexts

	cliticized	isolated argumental	phrased argumental	isolated adverbial	phrased adverbial
	kinoO=hodo	*kinoO*	*kinoO‿eraNda*	*kinoO*	*kinoO‿eraNda*
= UR	❶❶▼☾=❶	❶❶▼☾	❶❶▼☾ + ❶…	❶❶▼☾	❶❶▼☾ + ❶…
after composition	❶❶▼=Ⓛ❶				
after Monotonic Lowering	❶❶▼=ⓁⓁ				
after BAD				❶❶▼	❶❶▼ + ❶…
after Defective Lowering = SR	❶❶▽=ⓁⓁ	❶❶▽☾	❶❶▽☾ + ❶…		
after T-Phrasing			❶❶▽‿Ⓛ…		❶❶▼‿❶…
after Floating Removal = PR		❶❶▽			

(40) Full derivation of /*asita* ❶❶🔺☾/ "tomorrow" in various contexts

	cliticized	isolated argumental	phrased argumental	isolated adverbial	phrased adverbial
	asita=hodo	*asita*	*asita‿erabu*	*asita*	*asita‿erabu*
= UR	❶❶🔺☾=❶	❶❶🔺☾	❶❶🔺☾ + ❶…	❶❶🔺☾	❶❶🔺☾ + ❶…
after composition	❶❶🔺=Ⓛ❶				
after Monotonic Lowering	❶❶🔺=ⓁⓁ				
after BAD				❶❶🔺	❶❶🔺 + ❶…
after Defective Lowering = SR					
after T-Phrasing			❶❶🔺‿Ⓛ…		❶❶🔺‿❶…
after Floating Removal = PR		❶❶🔺			

Finally, (41) is a summary of the existence of floating skeletons and floating tones in various wordforms/PoSs:

(41) Summary of the existence of floating skeletons and tones in various PoSs/wordforms

PoS	example	cliticized?	floating skeleton?	floating tone w/o floating skeleton?
verb	(ind./adn.) *arau* (infinitive) *arai* (*te*-form) *araQte*	optionally	no	[a]
i-adjective	(ind./adn.) *akaI* (infinitive) *akaku* (*te*-form) *akakute*	optionally	no	no
na-adjective	(*da*-ind.) *husigi-da* (adnominal) *husigi-na* (infinitive) *husigi-ni/de* (*te*-form) *husigi-de*	optionally	no	no
	(∅-ind.) *husigi*	**always**	**yes**	no
noun	*kinoO*	optionally	(cliticized) **yes** (bare-argumental) **yes** (bare-adverbial) no	no
adverb	*maQtaku*	optionally	no	[b]
particle	*nara*	optionally	no	[c]
adnominal	*kono*	**never**	no	no

[a] Only with accented monomoraic infinitives.

[b] Exceptionally.

[c] Only with ⟨=*hodo*⟩, optionally.

4 The birth of a floating skeleton: a historical perspective

Where do floating skeletons come from, where does the split between BAD and bare-argumental t-phrasing come from, and why are they there only after nominals? A brief historical introduction to Tokyo tonology may help to answer these questions. (For Middle Japanese accentuation data, see Martin 1987; for tone interpretation, see Ramsey 1979; de Boer 2010.)

4.1 Where do floating skeletons come from?

In Middle Japanese, clitic phonology was fairly transparent. Lexical bases and particles mostly had an unchanging realization throughout cliticization. Specifically:

- Unaccented nouns were invariantly all-[Ⓛ]; (42d)
- Final-accented nouns were invariantly final-[❶]; (42c)
- Monomoraic case particles like ⟨=*o*⟩ (=ACC), being the most typical (frequent and phonologically dependent) enclitics today, were invariantly [=Ⓛ] (see Kibe 1983: 31; Martin 1987: 170).

(42) Historical development of noun (+ ⟨=*o*⟩)

	base example	Modern Tokyo base accent class	Middle Japanese		Modern Tokyo SR	Modern Tokyo base UR	
(a)	*takasa*	initial-accented	3.6	❶ⓁⓁ(=Ⓛ)	❶ⓁⓁ(=Ⓛ)	❶ⓁⓁ◖	Monotonizing
	itigo		3.7b	❶Ⓛ❶(=Ⓛ)			
(b)	*kokoro*	medial-accented	3.5	❶❶Ⓛ(=Ⓛ)	❶❶Ⓛ(=Ⓛ)	❶❶Ⓛ◖	
(c)	*sasimi*	final-accented	3.4	❶❶❶(=Ⓛ)	❶❶❶(=Ⓛ)	❶❶❶◖	Monotonizing
	azuki		3.2b	ⓁⓁ❶(=Ⓛ)			
(d)	*katati*	unaccented	3.1	ⓁⓁⓁ(=Ⓛ)	❶❶❶(=❶)	❶❶❶◖	All-[Ⓛ] Raising

Then followed two major sound changes:

- **Monotonizing**: Identify the first phonemic fall of a t-word; raise all tones before the fall to ❶, and all tones after it to Ⓛ.
- **All-[Ⓛ] Raising**: Raise all-[Ⓛ] t-words to all-[❶]. (42d)

Regarding floating-genesis, All-[Ⓛ] Raising was the crucial sound change. After All-[Ⓛ] Raising, which raised the particle from [=Ⓛ] to [=❶] as in (42d), the particle started to show tonal alternation between [=Ⓛ] and [=❶]. This [=❶], conditional on the base, must be attributed to the underlying specification of the base. At that stage, trailing floating high tones (◖) after unaccented noun bases (42d) were born.

The UR imbalance brought by the new-born floating high tones after unaccented nouns, but not after accented nouns, later triggered an analogical change as shown in (43): on seeing that unaccented nouns are capable of forcing a tone on the following recessive particle, final-accented nouns felt like doing so, too. When the analogy happened, trailing floating low tones (◗) were born, and so were trailing floating skeletons (↾).

(43) Analogical change in final-accented nouns (model; target; change)

	unaccented	final-accented		
	itigo	*sasimi*		
N + /=*o* =Ⓛ/ (=ACC)	❶❶❶=❶	❶❶❶=Ⓛ	=	❶❶❶=Ⓛ
N + /=*to* =❶/ (=QUOT)	❶❶❶=❶	❶❶❶=❶	≫	❶❶❶=Ⓛ
base UR	❶❶❶◖	❶❶❶	≫	❶❶❶◗

4.2 Where does the split between BAD and bare-argumental t-phrasing come from?

Before the birth of floating skeletons, the original version of t-phrasing must have been as straightforward as our first approximation of it (i.e., through simple concatenation of t-words, without any complication at the juncture). Then, how did t-phrasing evolve into its modern version, with conditional bleeding by BAD? Note the following analogical change in (44):

(44) Analogical changes in t-phrasing patterns (model; target; change)

	after sound changes		modern	gloss	
	w/ case marker	w/o case marker			
(a)	❶❶-Ⓛ + ❶Ⓛ → *natu^{ʟ}=ga ki^{ʟ}ta*	❶❶ + ❶Ⓛ → *natu° ki^{ʟ}ta*	❶❶◜ + ❶Ⓛ → *natu^{ʟ} ki^{ʟ}ta*	summer(=NOM) + came “summer has come”	bare-arg’l t-phrase
(b)	❶❶-❶ + ❶Ⓛ → *baN°=ga ki^{ʟ}ta*	❶❶ + ❶Ⓛ → *baN° ki^{ʟ}ta*	❶❶◖ + ❶Ⓛ → *baN° ki^{ʟ}ta*	evening(=NOM) + came “evening has come”	
(c)	—	❶❶ + ❶Ⓛ → *natu° ki^{ʟ}ta*	❶❶ + ❶Ⓛ → *natu° ki^{ʟ}ta*	summer + came “(someone) came in summer”	general t-phrase (w/ BAD)
(d)	—	❶❶ + ❶Ⓛ → *baN° ki^{ʟ}ta*	❶❶ + ❶Ⓛ → *baN° ki^{ʟ}ta*	evening + came “(someone) came in the evening”	

By recovering the implicit case marker, a bare-argumental t-phrase can always be related to its corresponding general t-phrase counterpart. These two corresponding phrases freely interchange with each other throughout the history of the language. In that case, *⟦*natu° ki^{ʟ}ta*⟧ changed to ⟦*natu^{ʟ} ki^{ʟ}ta*⟧ by analogy with its general t-phrase counterpart, ⟦*natu^{ʟ}=ga ki^{ʟ}ta*⟧. The modern version of t-phrasing is then obtained through reanalysis.

In contrast, in bare adverbial t-phrases, a similar analogical change failed due to a lack of a corresponding form. The application of the original t-phrasing rule (i.e., surface concatenation) to bare adverbial t-phrases must now be captured by the special rule of BAD in the reanalysed system.

4.3 Why are floating skeletons there only after nominals?

Short answer: it suffices to note that, among the two inflected forms of predicates that readily enter into cliticization—the indicative and the infinitive—no wordforms were all-[Ⓛ] in Middle Japanese. Therefore, no form was subject to All-[Ⓛ] Raising.

The development of the indicative form of predicates is listed in (45):

- All indicatives were [...❶Ⓛ] for accented predicates, and [...Ⓛ❶] (> [...❶❶] through Monotonizing) for unaccented predicates.
- Again, monomoraic final particles like ⟨*=ka*⟩ (=Q) were invariantly [=Ⓛ], and have remained so today.

(45) Historical development of the indicative (+ ⟨*=ka*⟩) (◍$_1$: *SPE* notation)

	base example	base class	Middle J. SR	Modern Tokyo SR	Modern Tokyo base UR	
(a)	*sirosi* ≫ *siroI*	acc. *i*-adj.	❶$_1$Ⓛ(=Ⓛ)	❶$_1$▽(=Ⓛ)	❶$_1$Ⓛ	
(b)	*oyogu*	acc. verb	❶$_1$Ⓛ(=Ⓛ)	❶$_1$Ⓛ(=Ⓛ)	❶$_1$Ⓛ	
(c)	*akasi* ≫ *akaI*	unacc. *i*-adj.	Ⓛ$_1$❶(=Ⓛ)	❶$_1$▼ ❶$_1$▽=Ⓛ	❶$_1$❶	Monotonizing Defective Lowering
(d)	*arau*	unacc. verb	Ⓛ$_1$❶(=Ⓛ)	❶$_1$❶(=Ⓛ)	❶$_1$❶	Monotonizing

Therefore, a floating skeleton is unmotivated for the indicative of predicates, because there wasn't a ◖ introduced by All-[Ⓛ] Raising to start with.

The development of the infinitive form of predicates is listed in (46). The modern surface pattern and base URs can be similarly accounted for as we have done with indicative bases, and are left to the reader to verify. Note especially that, with unaccented adjectives, there's an innovative variant (e.g. [*akaku=wa* ⒽⒽⓁ=Ⓛ]), generated under the analogy with the *te*-form (e.g. [*akakute* ⒽⒽⓁⓁ]). (See § 3.7.)

(46) Historical development of the infinitive (+ ⟨=*wa*⟩) (◍$_1$: *SPE* notation)

	base example	base class	Middle J. SR	Modern Tokyo SR	Modern Tokyo base UR	
(a)	*siroku*	acc. *i*-adj.	❶$_1$Ⓛ❶(=Ⓛ)	❶$_1$ⓁⓁ(=Ⓛ)	❶$_1$ⓁⓁ	Monotonizing
(b)	*oyogu*	acc. verb	❶$_1$Ⓛ(=Ⓛ)	❶$_1$Ⓛ(=Ⓛ)	❶$_1$Ⓛ	
(c)	*akaku*	unacc. *i*-adj.	Ⓛ$_1$❶(=Ⓛ)	❶$_1$❶(=Ⓛ) ~ ❶$_1$Ⓛ=Ⓛ	❶$_1$❶ ~ ❶$_1$Ⓛ=	Monotonizing analogy with the *te*-form
(d)	*arau*	unacc. verb	Ⓛ$_1$❶(=Ⓛ)	❶$_1$❶(=Ⓛ)	❶$_1$❶	Monotonizing

5 Discussion

5.1 Ubiquitous floating tone × PoS effect

Admittedly, the floating tone analysis of Tokyo Japanese as advanced in this paper is typologically unusual, in having an entire PoS of lexical words possessing floating tones. Normally, floating tones as a deviation from the well-formedness constraints are exceptional and non-systematic. That said, it is not the first time systematic floating tones have been reported. According to Rolle (2026), the Nigerian language Izon is described as having ubiquitous floating tones: all lexical items in Izon necessarily fall into one of the 4 floating tone classes and require some floating tones. (47) lists the 4 tone classes of Izon, where the trailing tones of nouns are realized on the verb that follows when the noun and the verb are in the same tone group, as in the phrase ⟨… *ẹrị*⟩ "see …!":

(47) The 4 tone classes of Izon (Rolle 2026: 34)

class	floating tone	example					
		UR			… + /*ẹrị* ⓁⒽ◖/		
A	…◹◖	*ta*	Ⓛ◹◖	"wife"	*ta͡ẹrị*	Ⓛ͡ⓁⒽ	"see wife!"
B	…◖	*fụ*	Ⓗ◖	"salt"	*fụ͡ẹrị*	Ⓗ͡ⒽⒽ	"see salt!"
C	…◹	*wun*	Ⓗ◹	"sand"	*wun͡ẹrị*	Ⓗ͡ⓁⓁ	"see sand!"
D	…◖◹	*wo*	Ⓛ◖◹	"him"	*wo͡ẹrị*	Ⓛ͡ⒽⓁ	"see him!"

The difference here is that, in Tokyo tonology, the floating tone is ubiquitous only within certain PoSs, namely nouns and *na*-adjectives. Put otherwise, what is seen in Tokyo tonology is a combination of the floating tone ubiquity and the PoS effect. Smith (2012) is a survey of PoS effects in phonology, where she discusses the PoS hierarchy of phonological privilege. Drawing on the contrast patterns of docked tones in Tokyo Japanese (recall that accented nouns contrast for the accent position while accented verbs and *i*-adjectives don't, as shown in (4) and (5)) among others, the survey concludes that nouns are more phonologically privileged than adjectives and verbs in being able to sponsor more phonological contrasts. The very conclusion holds true of floating tones in Tokyo tonology as well: final-[Ⓗ] nouns contrast for accentedness (unaccented /*itigo* ⒽⒽⒽ◖/ vs. final-accented /*sasimi* ⒽⒽⒽ◹/) while final-[Ⓗ] verbs and *i*-adjectives don't. Put otherwise, being able to sponsor or even require a floating tone is another manifestation of nouns' phonological privilege.

As for the origin of PoS effects in phonology, Smith (2012) speculates that the effect may arise from a bias in language acquisition. As a general cross-linguistic tendency this may well be true. However, for Tokyo tonology specifically, we have learned from § 4 above that such a PoS effect in phonology may also arise, by sheer coincidence, from an originally uninteresting pattern through regular sound changes plus some minimal analogy within a single PoS.

5.2 Downstep, pitch accent, or level tone?

Recall that in § 1 I characterize the subphrasal tonology of Tokyo as a restricted downstep system allowing at most one downstep within each t-word, which, by virtue of its culminative nature, would be more commonly called a pitch accent system. However, in ensuing discussions I have developed this floating reanalysis based on level tones. This dual treatment of a single phonological system may seem unexpected to traditional views, which have often conceptualized the pitch accent and the tone as two phenomena that should be clearly distinguished. For example, Hayes (2009: 291–293), based on this distinction, classifies languages into "tone languages" and "pitch accent languages". Then a pertinent question is: how should the

distinction between the two concepts be reconciled?

It should be noted that mainstream modern phonological theories no longer treat tones and pitch accents as two separate things. The tone is generally defined as a melodic device to distinguish word(form)s[28], disregarding whether it satisfies the culminativity constraint or how sparse its lexical specification is[29]. According to these definitions, the so-called pitch accent systems are, strictly speaking, a subtype of tones (Hyman 2001; Yip 2002: 259; Gussenhoven 2004: 41). This is exactly why many practitioners of Japanese phonology (including M. E. Beckman, E. M. de Boer, M. M. Clark, S. Haraguchi, H. Kubozono, J. B. Pierrehumbert, W. J. Poser) have openly used the term "tone" for the Tokyo Japanese pitch accent. More tellingly, Hyman's (2006) typological survey of tones and pitch accents shows that features like culminativity and obligatoriness are better viewed as elements that individual tonal systems can "pick and choose", rather than as hard and fast classifying criteria. In that sense, a pitch accent / downstep system is but a phonologically restricted tone system.

In previous sections I have shown that pitch accent analyses of Tokyo Japanese fail whilst my floating tone analysis works. What is left unexplained is the fundamental structural difference between the two analyses. Here is the crucial point: the high floating tone's override of the low tone on the first mora of the particle cannot be readily translated into the language of pitch accent analyses. To illustrate, alternative pitch accent analyses of ⟨*tamago=nara*⟩ directly translated from their floating tone counterpart are given in (48):

(48) Floating tone analysis of ⟨*tamago=nara*⟩ translated into pitch accent analyses

floating tone analysis	❽❽❽◖ + =Ⓛ Ⓛ	→	❽❽❽◖⃠Ⓛ	→	❽❽❽=❽Ⓛ (= ◍◍◍=◍◍)	✓
alternative PA analysis 1	◍◍◍◌ + =◍◍	→	◍◍◍=◍◍	→	◍◍◍=◍◍ (= ❽❽❽=ⓁⓁ)	✗
alternative PA analysis 2	◍◍◍◌ + =◍◍	→	◍◍◍=◍◍	=	◍◍◍=◍◍ (= ❽❽❽=ⓁⓁ)	✗

/=*nara* =ⓁⓁ/ is translated into /=◍◍/ in a PA analysis, with a downstep at the juncture to contrast with /=*hodo* =❽❽/ (which would be /=◍◍/, without a downstep). As for /*tamago* ❽❽❽◖/, there are two reasonable translations, either /◍◍◍◌/ or /◍◍◍◌/. However, in either case, a composition of the base and the particle would eventually yield [◍◍◍=◍◍] (= [❽❽❽=ⓁⓁ])—a wrong SR with the downstep inherited from /=*nara* =◍◍/ just at the juncture. The wrong prediction results from the fact that the downstep contributed by the particle /=*nara*/ cannot be cancelled by some usual device in pitch accent analyses. This is starkly different from the floating tone analysis, where the initial /Ⓛ/ of the particle /=*nara* =ⓁⓁ/ getting overridden by the floating /◖/ of /*tamago* ❽❽❽◖/ hardly strikes us as anything unusual.

To sum up, my reanalysis of Tokyo recessive t-cliticization shows that Tokyo tonology as a pitch accent system at the surface level may not be reduced to a pitch accent system underlyingly. More generally, a restricted phonological system at the surface level doesn't entail the same degree of phonological

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[28] ‣ "**tone language**: A language in which pitch differences function to differentiate words." (Carr 2009: 177)
‣ "Such languages [as Beijing Chinese], where word meanings or grammatical categories (such as tense) are dependent on pitch level, are known as tone languages." (Crystal 2008: 486, "tone")
‣ "In other languages, called tone languages, different pitch patterns are used in the same way vowels and consonants are used in all languages, i.e. to distinguish words from one another." (Gussenhoven & Jacobs 2011: 23)
‣ "**tone**: n. 1. The phenomenon, occurring in many languages, in which words of different meaning which consist of identical sequences of consonants and vowels are distinguished merely by contrasts of pitch." (Trask 1996: 356)
‣ "A language is a 'tone language' if the pitch of the word can change the meaning of the word." (Yip 2002: 1)

[29] A notable early exception is Pike's (1948: 3) definition of the tone: "A tone language may be defined as a language having lexically significant, contrastive, but relative pitch on each syllable." Note that, contrary to his classification, Beijing Mandarin is strictly speaking no longer a tone language in his definition because of the existence of Beijing's "neutral tone", which is commonly analysed as lexically unspecified for the tone and thus noncontrastive.

restrictedness at a deeper level.

On a final note, it has been commonplace to illustrate pitch accent systems using Tokyo Japanese's "(n+1)-way contrast on an n-mora word" (like ⟨*baꜜnana*⟩ : ⟨*tamaꜜgo*⟩ : ⟨*sasimiꜛ*⟩ : ⟨*itigo°*⟩ or so) in introductory texts. This is strictly speaking a confusion of surface and underlying analyses, and is better avoided. On the surface level, ⟨*sasimiꜛ*⟩ and ⟨*itigo°*⟩ share the same representation of [❶❶❶] and don't contrast; underlyingly, Tokyo Japanese cannot be reduced to a pitch accent system. Instead, one may say that Tokyo surface tonology is a pitch accent system that can accommodate at most an n-way contrast on an n-mora word, like [❶Ⓛ Ⓛ] (*banana*) : [❶❶Ⓛ] (*tamago*) : [❶❶❶] (*sasimi*, *itigo*).

6 Conclusion

In this paper, I argue that, as in the standard analysis and its existing alternatives to postlexical tonology of Tokyo Japanese, assuming a uniform downstep structure across different PoSs is a dead end. I note the Nominal Overriding Generalization that nominal (= noun and *na*-adjective) bases, and only these bases, always dictate a certain tone on the first mora of a following recessive particle. By coding this generalization as a floating skeleton that requires a floating tone at the end of these bases, my reanalysis successfully avoids the difficulties in previous analyses, without making unwarranted stipulations. I have also demonstrated that the split between bases with and without floating skeletons is rooted in the tonological history of Tokyo Japanese. In particular, high floating tones were born after all-[Ⓛ] t-words were raised to all-[Ⓗ], and low floating tones, along with floating skeletons, were created analogically under the influence of high floating tones.

The broader lessons are twofold. First, the case study enriches the typology of floating tones: unlike Izon, where floating tones are ubiquitous across all lexical items, Tokyo instantiates a PoS-restricted ubiquity, consistent with the phonological privilege of nouns. Second, Tokyo's surface pitch accent system is underlain by level tones, showing that a restricted surface system need not be underlain by an equally restricted system.

Tokyo Japanese tonology is one of the most well-studied pitch accent systems, and recessive t-cliticization encompasses the most productive and least idiosyncratic part of it. With such a fundamental reanalysis of the recessive t-cliticization of Tokyo Japanese, much of our knowledge of Japanese tonology and pitch accent systems in general needs to be re-examined.

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